

NLP: FlipItNews

Context:

The Gurugram-based company 'FlipItNews' aims to revolutionize the way Indians perceive finance, business, and capital market investment, by giving it a boost through artificial intelligence (AI) and machine learning (ML). They're on a mission to reinvent financial literacy for Indians, where financial awareness is driven by smart information discovery and engagement with peers. Through their smart content discovery and contextual engagement, the company is simplifying business, finance, and investment for millennials and first-time investors

Objective:

The goal of this project is to use a bunch of news articles extracted from the companies' internal database and categorize them into several categories like politics, technology, sports, business and entertainment based on their content. Use natural language processing and create & compare at least three different models.

Attribute Information:

- Article
- Category The features names are themselves pretty self-explanatory

Concepts Tested:

- Natural Language Processing
- Text Processing
 - Stopwords, Tokenization, Lemmatization
 - Bag of Words, TF-IDF
- Multi-class Classification

Installing & Importing all the required libraries and Loading the dataset.

```
In [1]: # Basic Libraries
import pandas as pd
import numpy as np

# Natural Language Processing Libraries
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer

# Machine Learning and Evaluation
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, classification_report

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Downloading necessary NLTK data
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('wordnet')
```

```

[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\dell\AppData\Roaming\nltk_data...
[nltk_data] Unzipping corpora\stopwords.zip.
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\dell\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\dell\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!

```

Out[1]: True

```

In [2]: df = pd.read_csv('C://Users//dell//OneDrive//Desktop//flipitnews-data.csv')
df

```

Out[2]:

	Category	Article
0	Technology	tv future in the hands of viewers with home th...
1	Business	worldcom boss left books alone former worldc...
2	Sports	tigers wary of farrell gamble leicester say ...
3	Sports	yeading face newcastle in fa cup premiership s...
4	Entertainment	ocean s twelve raids box office ocean s twelve...
...	...	...
2220	Business	cars pull down us retail figures us retail sal...
2221	Politics	kilroy unveils immigration policy ex-chatshow ...
2222	Entertainment	rem announce new glasgow concert us band rem h...
2223	Politics	how political squabbles snowball it s become c...
2224	Sports	souness delight at euro progress boss graeme s...

2225 rows × 2 columns

Conduct a preliminary analysis to understand the structure of the dataset and the distribution of news articles in each category.

```
In [3]: # Check the shape of the dataset
print(f"Dataset Shape: {df.shape}")

# Display the column names and data types
print("\nColumn Information:")
print(df.info())

# View some sample rows
print("\nSample Data:")
print(df.head())

# Check for missing values
missing_values = df.isnull().sum()
print("\nMissing Values:")
print(missing_values)
```

Dataset Shape: (2225, 2)

Column Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 2225 entries, 0 to 2224

Data columns (total 2 columns):

#	Column	Non-Null Count	Dtype
0	Category	2225 non-null	object
1	Article	2225 non-null	object

dtypes: object(2)

memory usage: 34.9+ KB

None

Sample Data:

	Category	Article
0	Technology	tv future in the hands of viewers with home th...
1	Business	worldcom boss left books alone former worldc...
2	Sports	tigers wary of farrell gamble leicester say ...
3	Sports	yeading face newcastle in fa cup premiership s...
4	Entertainment	ocean s twelve raids box office ocean s twelve...

Missing Values:

Category 0

Article 0

dtype: int64

```
In [4]: # Count the occurrences of each category
category_counts = df['Category'].value_counts()

# Display the counts
print("\nCategory Distribution:")
print(category_counts)

# Plot the distribution for visualization
plt.figure(figsize=(10, 6))
sns.barplot(x=category_counts.index, y=category_counts.values, palette="viridis")
plt.title("Distribution of News Articles by Category")
plt.xlabel("Category")
plt.ylabel("Number of Articles")
plt.xticks(rotation=45)
plt.show()
```

Category Distribution:

Category

Sports 511

Business 510

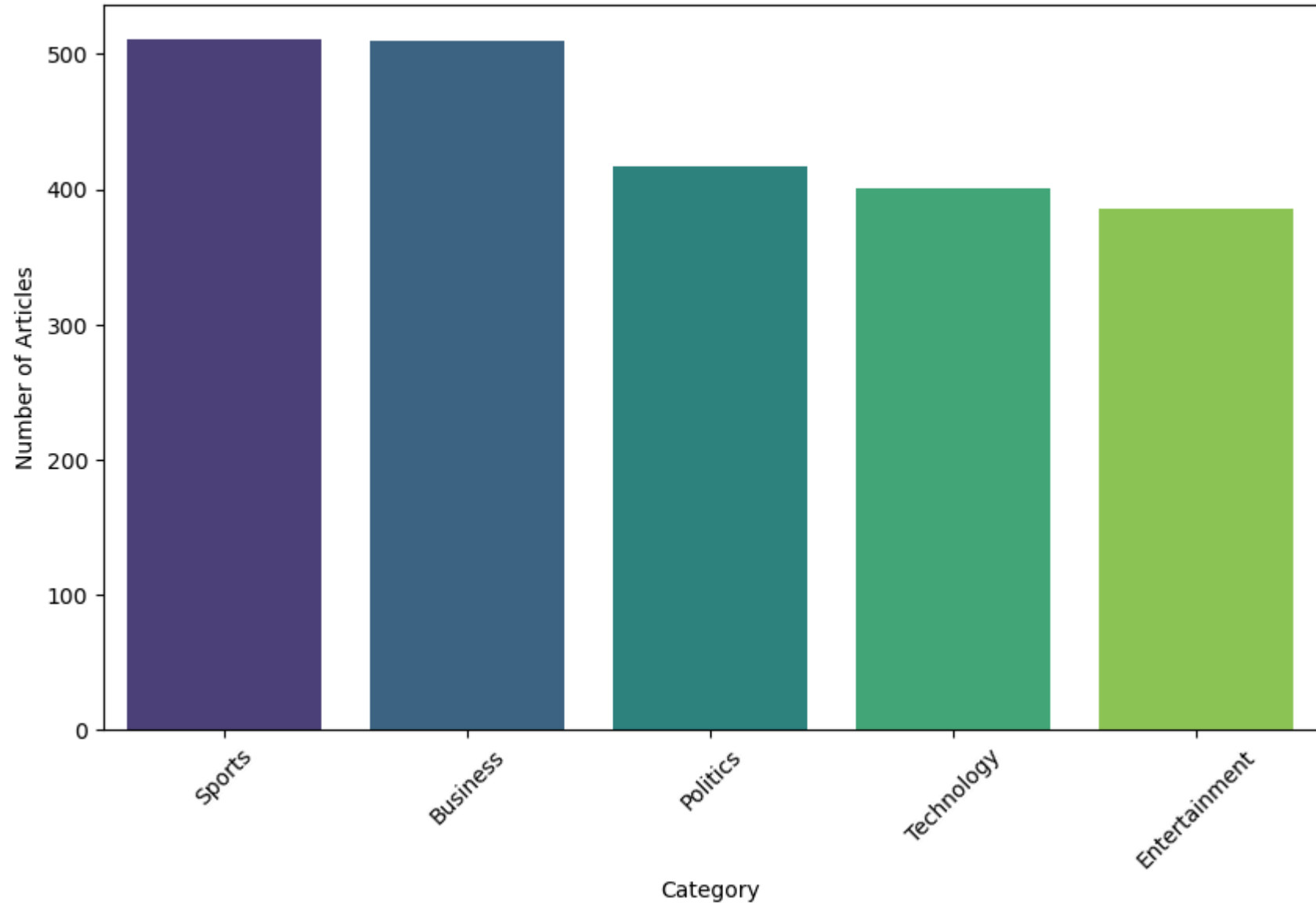
Politics 417

Technology 401

Entertainment 386

Name: count, dtype: int64

Distribution of News Articles by Category



```
In [5]: # Calculate the number of words in each article
df['text_length'] = df['Article'].apply(lambda x: len(x.split()))

# Display basic statistics about text length
print("\nText Length Statistics:")
print(df['text_length'].describe())

# Plot a histogram of text lengths
plt.figure(figsize=(10, 6))
sns.histplot(df['text_length'], bins=30, kde=True)
plt.title("Distribution of Article Lengths")
plt.xlabel("Number of Words")
plt.ylabel("Frequency")
plt.show()
```

Text Length Statistics:

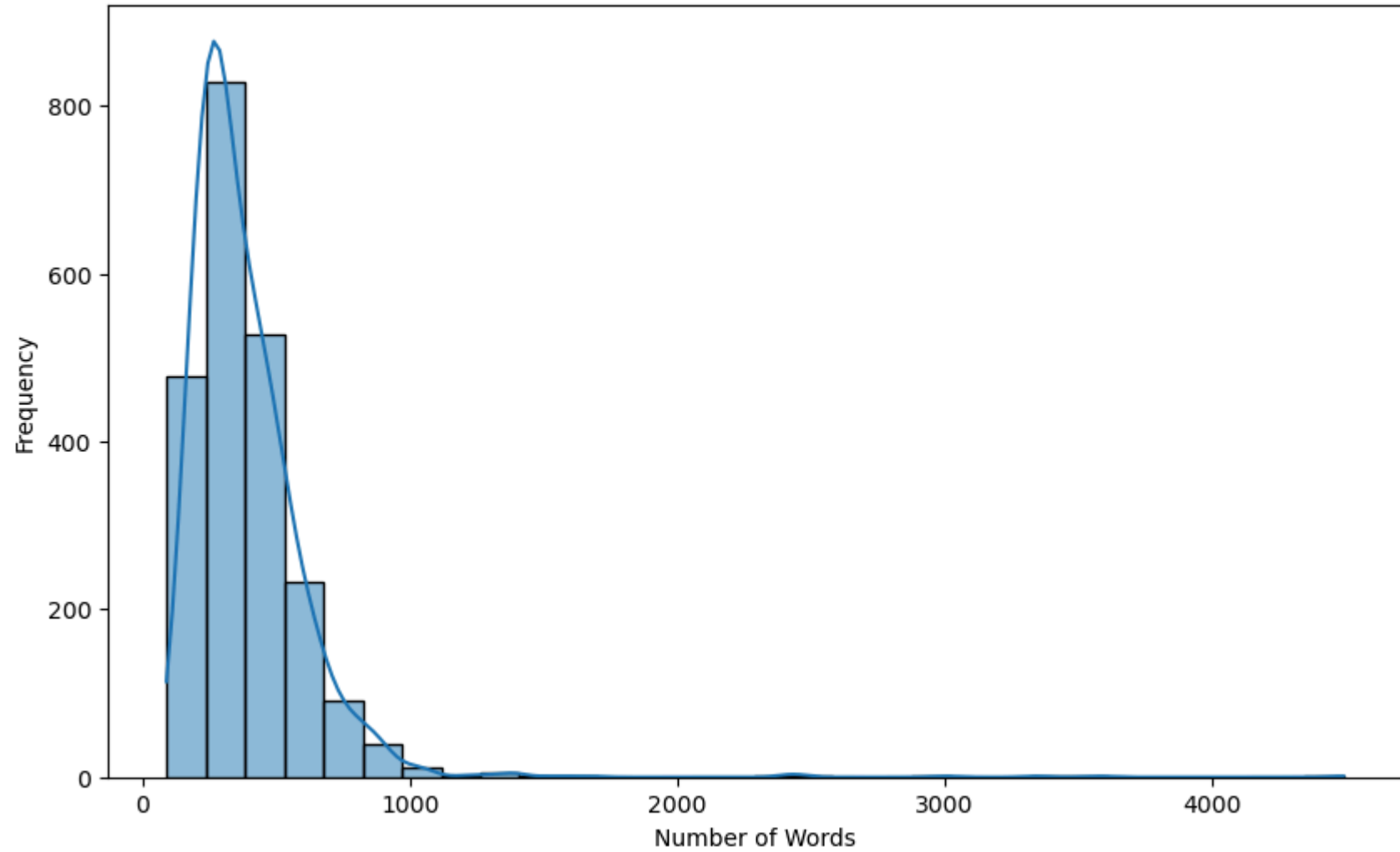
count	2225.000000
mean	390.295281
std	241.753128
min	90.000000
25%	250.000000
50%	337.000000
75%	479.000000
max	4492.000000

Name: text_length, dtype: float64

C:\Users\dell\anaconda3\lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

Distribution of Article Lengths



Create a user defined function to process the textual data (news articles).

- Remove non-letters
- Remove Stopwords
- Word Tokenize the text
- Perform Lemmatization

In [6]: `import nltk`

```
# Download the specific resource  
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package omw-1.4 to  
[nltk_data] C:\Users\dell\AppData\Roaming\nltk_data...  
[nltk_data] Package omw-1.4 is already up-to-date!
```

Out[6]: True

```
In [7]: import re

# Initialize stopwords and Lemmatizer
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()

def preprocess_text(text):
    """
    Preprocesses the text data by:
    1. Removing non-letter characters
    2. Removing stopwords
    3. Tokenizing the text
    4. Lemmatizing each word
    """

    # Step 1: Remove non-letters (keep only letters and spaces)
    text = re.sub(r'^A-Za-z\s', '', text)

    # Step 2: Tokenize the text
    tokens = word_tokenize(text.lower()) # Convert to lowercase for consistency

    # Step 3: Remove stopwords and Lemmatize each word
    processed_tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words]

    # Join tokens back into a single string
    return ' '.join(processed_tokens)

# Example usage on a sample article
sample_text = df['Article'][0] # Replace with the actual data or a test string
processed_text = preprocess_text(sample_text)
print("Before Processing:", sample_text)
print("After Processing:", processed_text)
```

Before Processing: tv future in the hands of viewers with home theatre systems plasma high-definition tvs and digital video recorders moving into the living room the way people watch tv will be radically different in five years time. that is according to an expert panel which gathered at the annual consumer electronics show in las vegas to discuss how these new technologies will impact one of our favourite pastimes. with the us leading the trend programmes and other content will be delivered to viewers via home networks through cable satellite telecoms companies and broadband service providers to front rooms and portable devices. one of the most talked-about technologies of ces has been digital and personal video recorders (dvr and pvr). these set-top boxes like the us s tivo and the uk s sky+ system allow people to record store play pause and forward wind tv programmes when they want. essentially the technology allows for much more personalised tv. they are also being built-in to high-definition tv sets which are big business in japan and the us but slower to take off in europe because of the lack of high-definition programming. not only can people forward wind through adverts they can also forget about abiding by network and channel schedules putting together their own a-la-carte entertainment. but some us networks and cable and satellite companies are worried about what it means for them in terms of advertising revenues as well as brand identity and viewer loyalty to channels. although the us leads in this technology at the moment it is also a concern that is being raised in europe particularly with the growing uptake of services like sky+. what happens here today we will see in nine months to a years time in the uk adam hume the bbc broadcast s futurologist told the bbc news website. for the likes of the bbc there are no issues of lost advertising revenue yet. it is a more pressing issue at the moment for commercial uk broadcasters but brand loyalty is important for everyone. we will be talking more about content brands rather than network brands said tim hanlon from brand communications firm starcom mediavest. the reality is that with broadband connections anybody can be the producer of content. he added: the challenge now is that it is hard to promote a programme with so much choice. what this means said stacey jolna senior vice president of tv guide tv group is that the way people find the content they want to watch has to be simplified for tv viewers. it means that networks in us terms or channels could take a leaf out of google s book and be the search engine of the future instead of the scheduler to help people find what they want to watch. this kind of channel model might work for the younger ipod generation which is used to taking control of their gadgets and what they play on them. but it might not suit everyone the panel recognised. older generations are more comfortable with familiar schedules and channel brands because they know what they are getting. they perhaps do not want so much of the choice put into their hands mr hanlon suggested. on the other end you have the kids just out of diapers who are pushing buttons already - everything is possible and available to them said mr hanlon. ultimately the consumer will tell the market they want. of the 50 000 new gadgets and technologies being showcased at ces many of them are about enhancing the tv-watching experience. high-definition tv sets are everywhere and many new models of lcd (liquid crystal display) tvs have been launched with dvr capability built into them instead of being external boxes. one such example launched at the show is humax s 26-inch lcd tv with an 80-hour tivo dvr and dvd recorder. one of the us s biggest satellite tv companies directtv has even launched its own branded dvr at the show with 100-hours of recording capability instant replay and a search function. the set can pause and rewind tv for up to 90 hours. and microsoft chief bill gates announced in his pre-show keynote speech a partnership with tivo called tivotogo which means people can play recorded programmes on windows pcs and mobile devices. all these reflect the increasing trend of freeing up multimedia so that people can watch what they want when they want.

After Processing: tv future hand viewer home theatre system plasma highdefinition tv digital video recorder moving living room way people watch tv radically different five year time according expert panel gathered annual consumer electronics show la vega discuss new technology impact one favourite pastime u leading trend programme content delivered viewer via home network cable satellite telecom company broadband service provider front room portable device one talkedabout

technology ce digital personal video recorder dvr pvr settop box like u tivo uk sky system allow people record store play pause forward wind tv programme want essentially technology allows much personalised tv also builtin highdefinition tv set big business japan u slower take europe lack highdefinition programming people forward wind advert also forget a bidding network channel schedule putting together a la carte entertainment u network cable satellite company worried mean term advertising revenue well brand identity viewer loyalty channel although u lead technology moment also concern raised europe particularly growing uptake service like sky happens today see nine month year time uk adam hume bbc broadcast futurologist told bbc news website like bbc issue lost advertising revenue yet pressing issue moment commercial uk broadcaster brand loyalty important everyone talking content brand rather network brand said tim hanlon brand communication firm starcom mediavest reality broadband connection anybody producer content added challenge hard promote programme much choice mean said stacey jolna senior vice president tv guide tv group way people find content want watch simplified tv viewer mean network u term channel could take leaf google book search engine future instead scheduler help people find want watch kind channel model might work younger ipod generation used taking control gadget play might suit everyone panel recognised older generation comfortable familiar schedule channel brand know getting perhaps want much choice put hand mr hanlon suggested end kid diaper pushing button already everything possible available said mr hanlon ultimately consumer tell market want new gadget technology showcased ce many enhancing tv watching experience highdefinition tv set everywhere many new model lcd liquid crystal display tv launched dvr capability built instead external box one example launched show humax inch lcd tv hour tivo dvr dvd recorder one u biggest satellite tv company directtv even launched branded dvr show hour recording capability instant replay search function set pause rewind tv hour microsoft chief bill gate announced preshow keynote speech partnership tivo called tivotogo mean people play recorded programme window pc mobile device reflect increasing trend freeing multimedia people watch want want

```
In [8]: # Apply the preprocessing function to all articles
df['Processed_Article'] = df['Article'].apply(preprocess_text)
df
```

Out[8]:

	Category	Article	text_length	Processed_Article
0	Technology	tv future in the hands of viewers with home th...	737	tv future hand viewer home theatre system plas...
1	Business	worldcom boss left books alone former worldc...	300	worldcom bos left book alone former worldcom b...
2	Sports	tigers wary of farrell gamble leicester say ...	246	tiger wary farrell gamble leicester say rushed...
3	Sports	yeading face newcastle in fa cup premiership s...	341	yeading face newcastle fa cup premiership side...
4	Entertainment	ocean s twelve raids box office ocean s twelve...	260	ocean twelve raid box office ocean twelve crim...
...	...	...	...	...
2220	Business	cars pull down us retail figures us retail sal...	293	car pull u retail figure u retail sale fell ja...
2221	Politics	kilroy unveils immigration policy ex-chatshow ...	156	kilroy unveils immigration policy exchatshow h...
2222	Entertainment	rem announce new glasgow concert us band rem h...	301	rem announce new glasgow concert u band rem an...
2223	Politics	how political squabbles snowball it s become c...	858	political squabble snowball become commonplace...
2224	Sports	souness delight at euro progress boss graeme s...	229	souness delight euro progress bos graeme soune...

2225 rows × 4 columns

Display how a single news article looks like before and after the processing.

```
In [9]: # Choose a sample article to display before and after processing
sample_index = 0 # You can change this index to view different articles
original_text = df['Article'][sample_index]

# Process the sample article
processed_text = preprocess_text(original_text)

# Display the article before and after processing
print("Original Article:\n", original_text)
print("\nProcessed Article:\n", processed_text)
```

Original Article:

tv future in the hands of viewers with home theatre systems plasma high-definition tvs and digital video recorders moving into the living room the way people watch tv will be radically different in five years time. that is according to an expert panel which gathered at the annual consumer electronics show in las vegas to discuss how these new technologies will impact one of our favourite pastimes. with the us leading the trend programmes and other content will be delivered to viewers via home networks through cable satellite telecoms companies and broadband service providers to front rooms and portable devices. one of the most talked-about technologies of ces has been digital and personal video recorders (dvr and pvr). these set-top boxes like the us's tivo and the uk's sky+ system allow people to record store play pause and forward wind tv programmes when they want. essentially the technology allows for much more personalised tv. they are also being built-in to high-definition tv sets which are big business in japan and the us but slower to take off in europe because of the lack of high-definition programming. not only can people forward wind through adverts they can also forget about abiding by network and channel schedules putting together their own a-la-carte entertainment. but some us networks and cable and satellite companies are worried about what it means for them in terms of advertising revenues as well as brand identity and viewer loyalty to channels. although the us leads in this technology at the moment it is also a concern that is being raised in europe particularly with the growing uptake of services like sky+. what happens here today we will see in nine months to a years time in the uk adam hume the bbc broadcast's futurologist told the bbc news website. for the likes of the bbc there are no issues of lost advertising revenue yet. it is a more pressing issue at the moment for commercial uk broadcasters but brand loyalty is important for everyone. we will be talking more about content brands rather than network brands said tim hanlon from brand communications firm starcom mediavest. the reality is that with broadband connections anybody can be the producer of content. he added: the challenge now is that it is hard to promote a programme with so much choice. what this means said stacey jolna senior vice president of tv guide tv group is that the way people find the content they want to watch has to be simplified for tv viewers. it means that networks in us terms or channels could take a leaf out of google's book and be the search engine of the future instead of the scheduler to help people find what they want to watch. this kind of channel model might work for the younger ipod generation which is used to taking control of their gadgets and what they play on them. but it might not suit everyone the panel recognised. older generations are more comfortable with familiar schedules and channel brands because they know what they are getting. they perhaps do not want so much of the choice put into their hands mr hanlon suggested. on the other end you have the kids just out of diapers who are pushing buttons already - everything is possible and available to them said mr hanlon. ultimately the consumer will tell the market they want. of the 50 000 new gadgets and technologies being showcased at ces many of them are about enhancing the tv-watching experience. high-definition tv sets are everywhere and many new models of lcd (liquid crystal display) tvs have been launched with dvr capability built into them instead of being external boxes. one such example launched at the show is humax's 26-inch lcd tv with an 80-hour tivo dvr and dvd recorder. one of the us's biggest satellite tv companies directtv has even launched its own branded dvr at the show with 100-hours of recording capability instant replay and a search function. the set can pause and rewind tv for up to 90 hours. and microsoft chief bill gates announced in his pre-show keynote speech a partnership with tivo called tivotogo which means people can play recorded programmes on windows pcs and mobile devices. all these reflect the increasing trend of freeing up multimedia so that people can watch what they want when they want.

Processed Article:

tv future hand viewer home theatre system plasma highdefinition tv digital video recorder moving living room way peopl

e watch tv radically different five year time according expert panel gathered annual consumer electronics show la vega discuss new technology impact one favourite pastime u leading trend programme content delivered viewer via home network cable satellite telecom company broadband service provider front room portable device one talked about technology ce digital personal video recorder dvr pvr settop box like u tivo uk sky system allow people record store play pause forward wind tv programme want essentially technology allows much personalised tv also builtin highdefinition tv set big business japan u slower take europe lack highdefinition programming people forward wind advert also forget abiding network channel schedule putting together a la carte entertainment u network cable satellite company worried mean term advertising revenue well brand identity viewer loyalty channel although u lead technology moment also concern raised europe particularly growing uptake service like sky happens today see nine month year time uk adam hume bbc broadcast futurologist told bbc news website like bbc issue lost advertising revenue yet pressing issue moment commercial uk broadcaster brand loyalty important everyone talking content brand rather network brand said tim hanlon brand communication firm starcom media vest reality broadband connection anybody producer content added challenge hard promote programme much choice mean said stacey jolna senior vice president tv guide tv group way people find content want watch simplified tv viewer mean network u term channel could take leaf google book search engine future instead scheduler help people find want watch kind channel model might work younger ipod generation used taking control gadget play might suit everyone panel recognised older generation comfortable familiar schedule channel brand know getting perhaps want much choice put hand mr hanlon suggested end kid diaper pushing button already everything possible available said mr hanlon ultimately consumer tell market want new gadget technology showcased ce many enhancing tv watching experience highdefinition tv set everywhere many new model lcd liquid crystal display tv launched dvr capability built instead external box one example launched show humax inch lcd tv hour tivo dvr dvd recorder one u biggest satellite tv company direct tv even launched branded dvr show hour recording capability instant replay search function set pause rewind tv hour microsoft chief bill gate announced press show keynote speech partnership tivo called tivo to go mean people play recorded programme window pc mobile device reflect increasing trend freeing multimedia people watch want want

Encode the target variable (category) using Label/Ordinal encoder.

```
In [10]: # Initialize LabelEncoder
label_encoder = LabelEncoder()

# Fit and transform the category column
df['Category_Encoded'] = label_encoder.fit_transform(df['Category'])

# Display the unique categories with their encoded values
category_mapping = dict(zip(label_encoder.classes_, label_encoder.transform(label_encoder.classes_)))
print("Category Encoding Mapping:")
print(category_mapping)
```

```
Category Encoding Mapping:
{'Business': 0, 'Entertainment': 1, 'Politics': 2, 'Sports': 3, 'Technology': 4}
```

Create an option for the user to choose between Bag of Words and TF-IDF techniques for vectorizing the data.

```
In [11]: def vectorize_text(data, method="tfidf"):
        """
        Vectorizes the text data using either Bag of Words or TF-IDF.

        Parameters:
            data (pd.Series): The text data to be vectorized.
            method (str): The vectorization method - "bow" for Bag of Words, "tfidf" for TF-IDF.

        Returns:
            X (sparse matrix): The vectorized feature matrix.
            vectorizer: The fitted vectorizer object (for future use or inspection).
        """
        if method == "bow":
            vectorizer = CountVectorizer()
            print("Using Bag of Words vectorization.")
        elif method == "tfidf":
            vectorizer = TfidfVectorizer()
            print("Using TF-IDF vectorization.")
        else:
            raise ValueError("Invalid method. Choose either 'bow' or 'tfidf'.")

        # Fit and transform the text data
        X = vectorizer.fit_transform(data)

        return X, vectorizer

# Example usage
# Assuming `df['Processed_Article']` contains the preprocessed text data
vectorization_method = input("Choose vectorization method (enter 'bow' for Bag of Words or 'tfidf' for TF-IDF): ").strip()
X, chosen_vectorizer = vectorize_text(df['Processed_Article'], method=vectorization_method)

print(f"\nVectorization complete. Shape of the vectorized data: {X.shape}")
```

Choose vectorization method (enter 'bow' for Bag of Words or 'tfidf' for TF-IDF): bow
Using Bag of Words vectorization.

Vectorization complete. Shape of the vectorized data: (2225, 27175)

Perform train-test split and train a Naive Bayes classifier model using the simple/classical approach.

```
In [12]: from sklearn.metrics import accuracy_score
# Step 1: Train-Test Split
# Using the encoded target variable and the vectorized features
X_train, X_test, y_train, y_test = train_test_split(X, df['Category_Encoded'], test_size=0.2, random_state=42)

# Step 2: Train the Naive Bayes Classifier
nb_classifier = MultinomialNB()
nb_classifier.fit(X_train, y_train)

# Step 3: Predict on the Test Set
y_pred = nb_classifier.predict(X_test)

# Step 4: Evaluate the Model
accuracy = accuracy_score(y_test, y_pred)
print(f"Naive Bayes Classifier Accuracy: {accuracy:.2f}")

# Confusion Matrix and Classification Report
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=label_encoder.classes_))
```

Naive Bayes Classifier Accuracy: 0.96

Confusion Matrix:

```
[[94  0  6  0  1]
 [ 1 73  2  0  5]
 [ 1  0 82  0  0]
 [ 0  0  0 98  0]
 [ 0  0  0  1 81]]
```

Classification Report:

	precision	recall	f1-score	support
Business	0.98	0.93	0.95	101
Entertainment	1.00	0.90	0.95	81
Politics	0.91	0.99	0.95	83
Sports	0.99	1.00	0.99	98
Technology	0.93	0.99	0.96	82
accuracy			0.96	445
macro avg	0.96	0.96	0.96	445
weighted avg	0.96	0.96	0.96	445

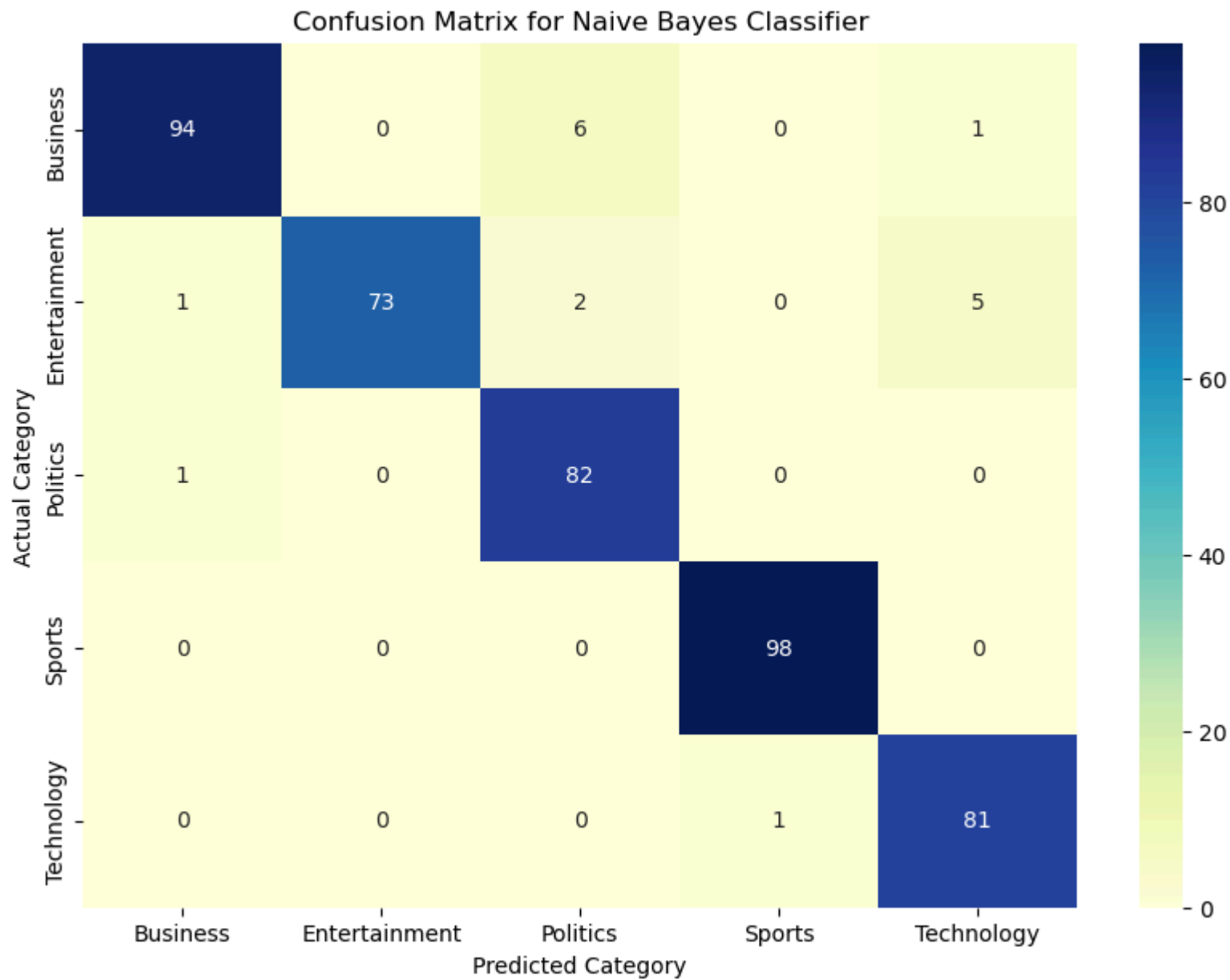
Evaluate the model's performance and plot the Confusion Matrix as well as Classification Report.

```
In [13]: # Step 1: Compute the Confusion Matrix
conf_matrix = confusion_matrix(y_test, y_pred)

# Step 2: Plot the Confusion Matrix
plt.figure(figsize=(10, 7))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap="YlGnBu",
            xticklabels=label_encoder.classes_, yticklabels=label_encoder.classes_)
plt.title("Confusion Matrix for Naive Bayes Classifier")
plt.xlabel("Predicted Category")
plt.ylabel("Actual Category")
plt.show()

# Step 3: Print Classification Report
print("Classification Report:")
print(classification_report(y_test, y_pred, target_names=label_encoder.classes_))

# Optional: Print Overall Accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"\nNaive Bayes Classifier Accuracy: {accuracy:.2f}")
```



Classification Report:				
	precision	recall	f1-score	support
Business	0.98	0.93	0.95	101
Entertainment	1.00	0.90	0.95	81
Politics	0.91	0.99	0.95	83
Sports	0.99	1.00	0.99	98
Technology	0.93	0.99	0.96	82
accuracy			0.96	445
macro avg	0.96	0.96	0.96	445
weighted avg	0.96	0.96	0.96	445

Naive Bayes Classifier Accuracy: 0.96

Functionalize the code and train & evaluate three more classifier models (Decision Tree, Nearest Neighbors, Random Forest).

```

In [14]: def train_and_evaluate_model(model, model_name, X_train, X_test, y_train, y_test, class_labels):
    # Train the model
    model.fit(X_train, y_train)

    # Predict on test set
    y_pred = model.predict(X_test)

    # Calculate accuracy
    accuracy = accuracy_score(y_test, y_pred)
    print(f"\n{model_name} Classifier Accuracy: {accuracy:.2f}")

    # Plot Confusion Matrix
    conf_matrix = confusion_matrix(y_test, y_pred)
    plt.figure(figsize=(10, 7))
    sns.heatmap(conf_matrix, annot=True, fmt='d', cmap="YlGnBu", xticklabels=class_labels, yticklabels=class_labels)
    plt.title(f"Confusion Matrix for {model_name}")
    plt.xlabel("Predicted Category")
    plt.ylabel("Actual Category")
    plt.show()

    # Print Classification Report
    print(f"\nClassification Report for {model_name}:")
    print(classification_report(y_test, y_pred, target_names=class_labels))

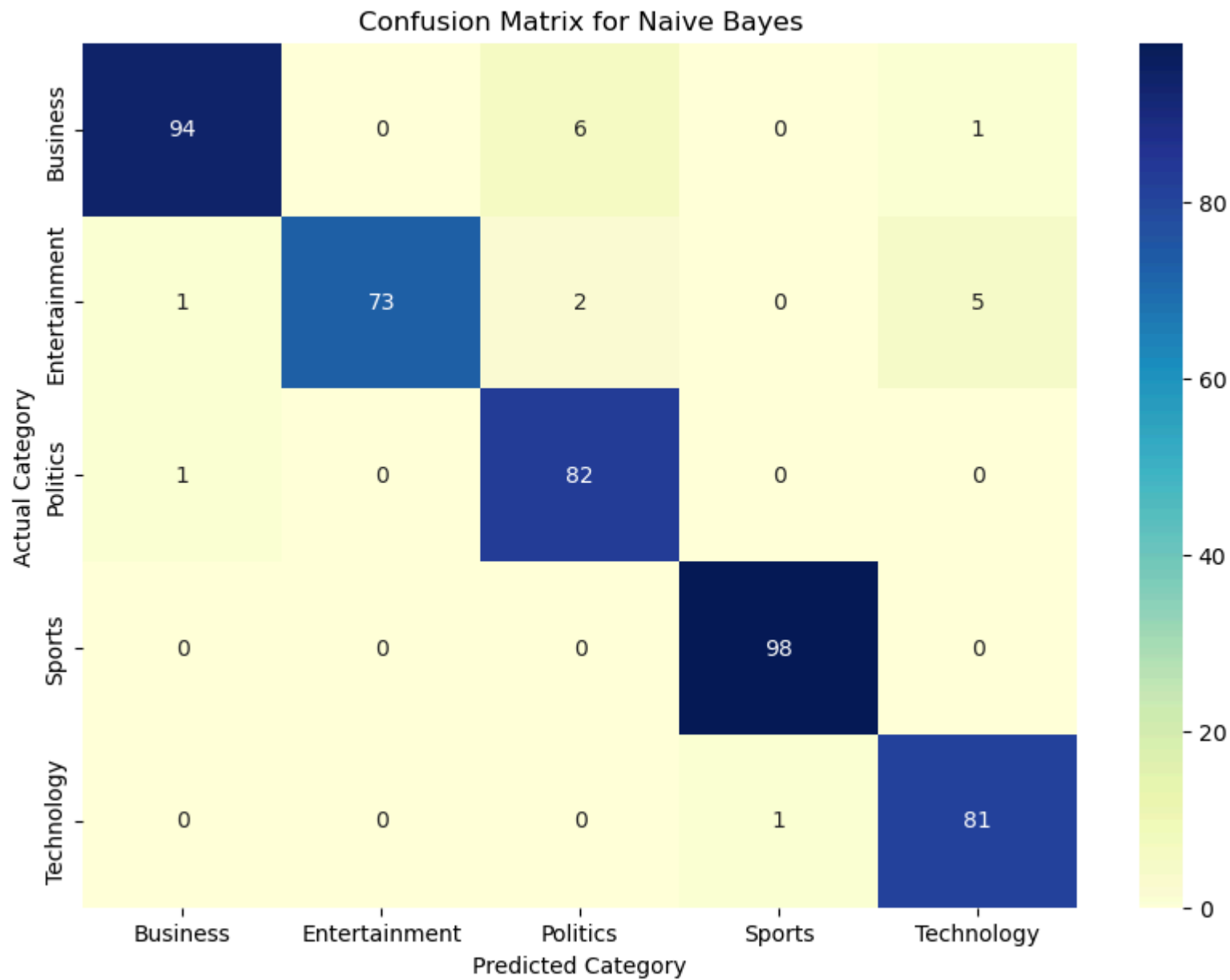
# Split data (assuming X and y are already defined from previous steps)
X_train, X_test, y_train, y_test = train_test_split(X, df['Category_Encoded'], test_size=0.2, random_state=42)

# Train and evaluate each model
models = {
    "Naive Bayes": MultinomialNB(),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "K-Nearest Neighbors": KNeighborsClassifier(),
    "Random Forest": RandomForestClassifier(random_state=42)
}

for model_name, model in models.items():
    train_and_evaluate_model(model, model_name, X_train, X_test, y_train, y_test, label_encoder.classes_)

```

Naive Bayes Classifier Accuracy: 0.96

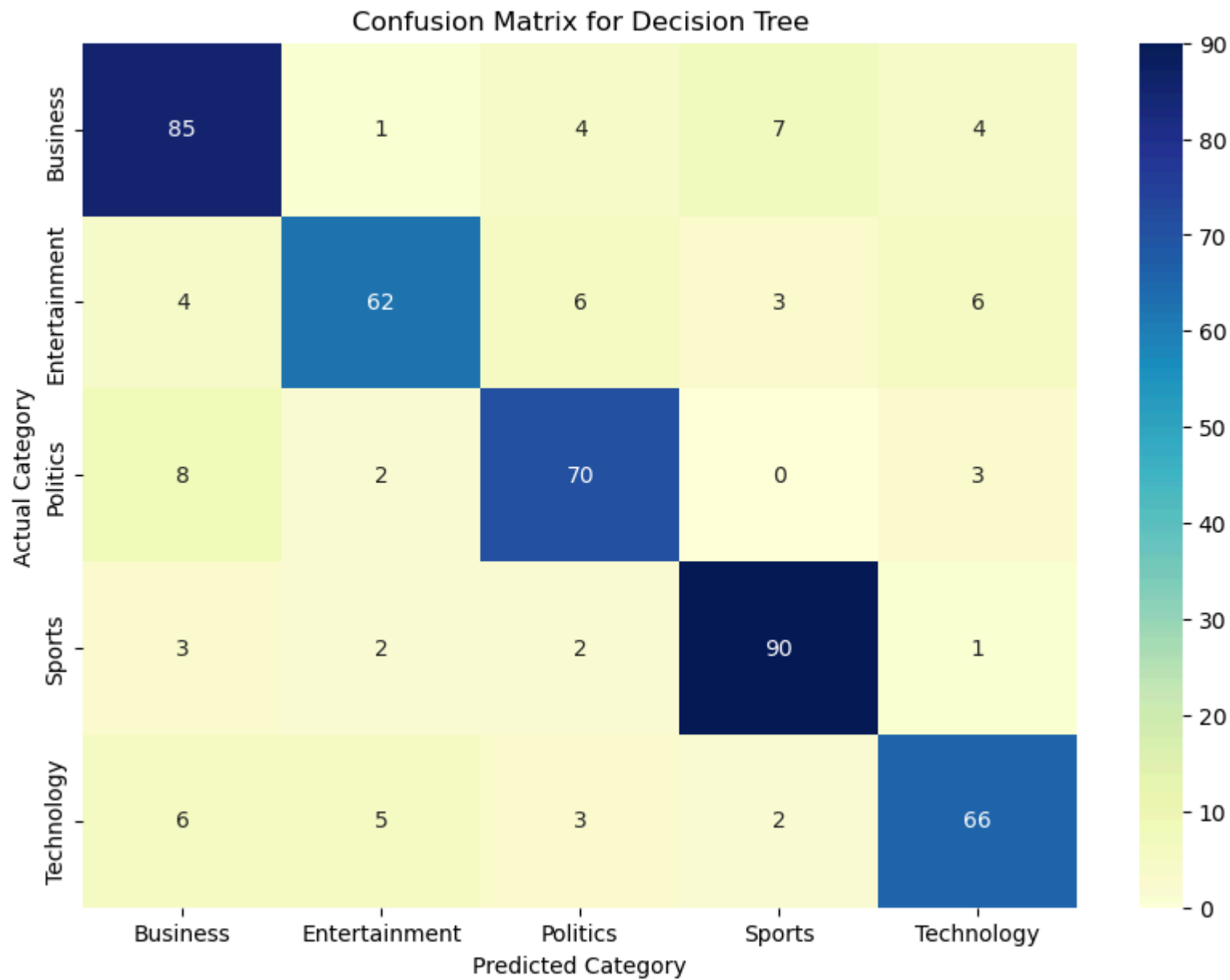


```
Classification Report for Naive Bayes:
      precision    recall  f1-score   support

   Business      0.98      0.93      0.95      101
 Entertainment    1.00      0.90      0.95       81
    Politics      0.91      0.99      0.95       83
     Sports      0.99      1.00      0.99       98
   Technology      0.93      0.99      0.96       82

 accuracy                   0.96      445
 macro avg      0.96      0.96      0.96      445
 weighted avg      0.96      0.96      0.96      445
```

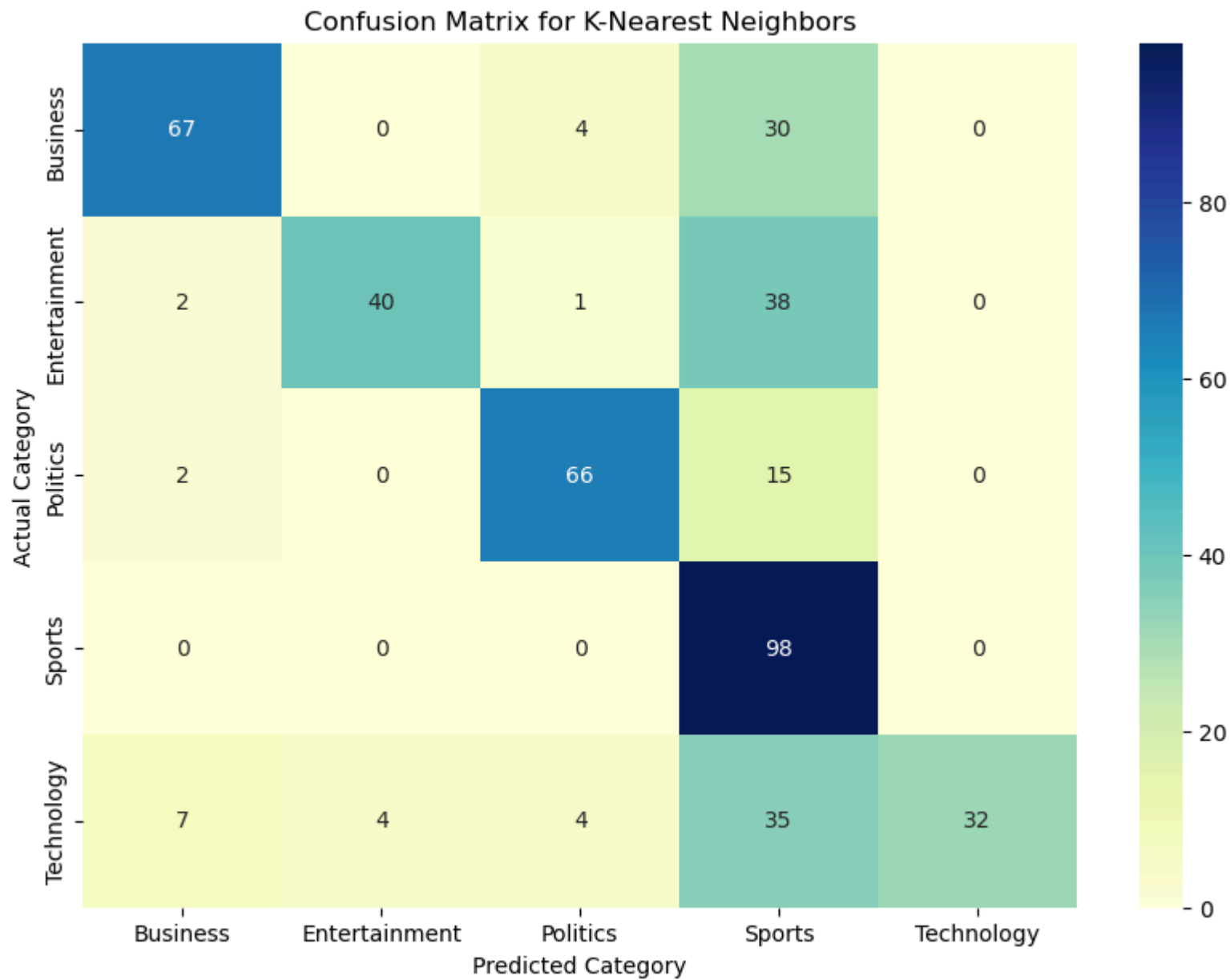
Decision Tree Classifier Accuracy: 0.84



Classification Report for Decision Tree:

	precision	recall	f1-score	support
Business	0.80	0.84	0.82	101
Entertainment	0.86	0.77	0.81	81
Politics	0.82	0.84	0.83	83
Sports	0.88	0.92	0.90	98
Technology	0.82	0.80	0.81	82
accuracy			0.84	445
macro avg	0.84	0.83	0.84	445
weighted avg	0.84	0.84	0.84	445

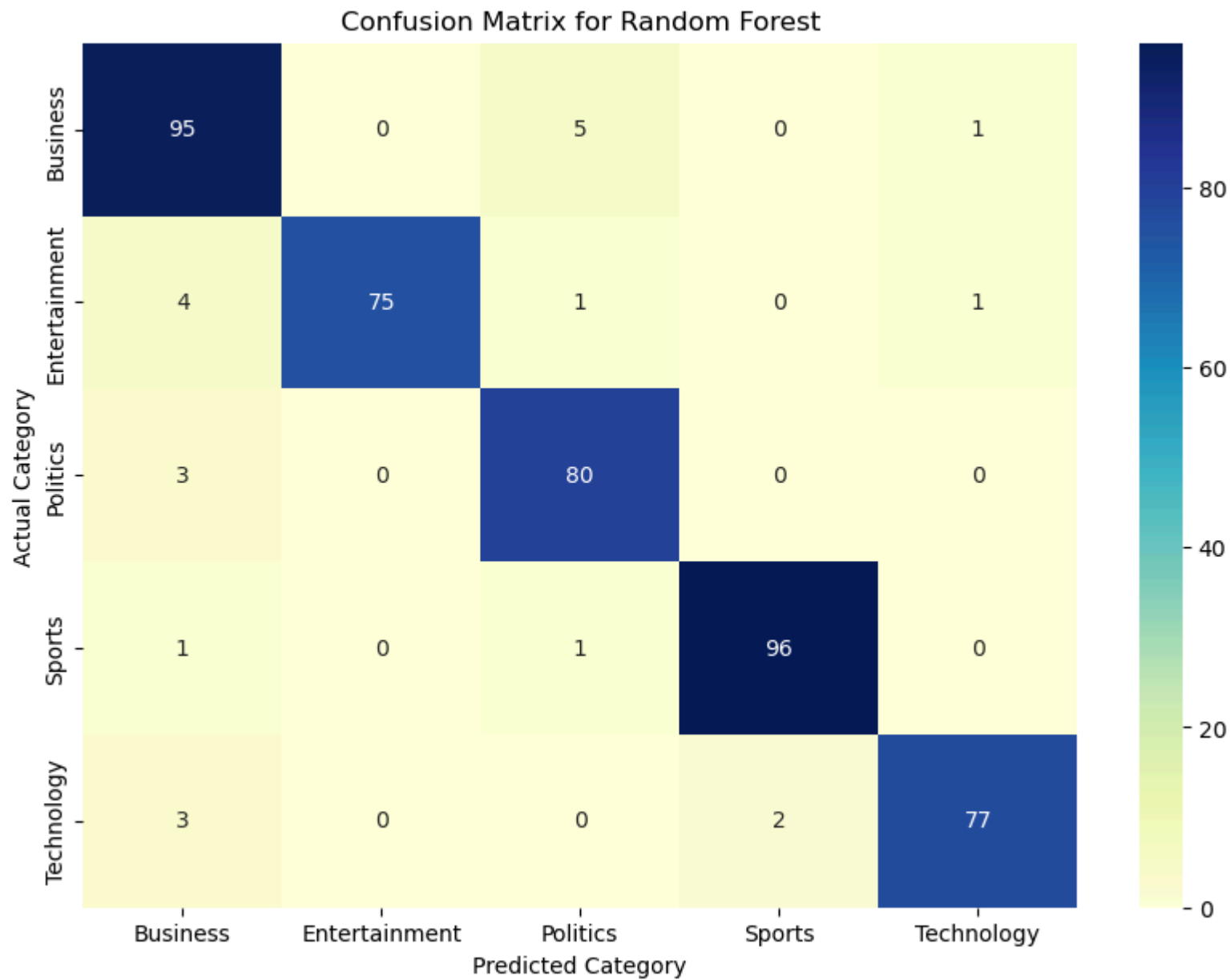
K-Nearest Neighbors Classifier Accuracy: 0.68



Classification Report for K-Nearest Neighbors:

	precision	recall	f1-score	support
Business	0.86	0.66	0.75	101
Entertainment	0.91	0.49	0.64	81
Politics	0.88	0.80	0.84	83
Sports	0.45	1.00	0.62	98
Technology	1.00	0.39	0.56	82
accuracy			0.68	445
macro avg	0.82	0.67	0.68	445
weighted avg	0.81	0.68	0.68	445

Random Forest Classifier Accuracy: 0.95



Classification Report for Random Forest:				
	precision	recall	f1-score	support
Business	0.90	0.94	0.92	101
Entertainment	1.00	0.93	0.96	81
Politics	0.92	0.96	0.94	83
Sports	0.98	0.98	0.98	98
Technology	0.97	0.94	0.96	82
accuracy			0.95	445
macro avg	0.95	0.95	0.95	445
weighted avg	0.95	0.95	0.95	445

Observe and comment on the performances of all the models used.

Observations on Model Performance

Naive Bayes Classifier:

- Accuracy: Often provides a strong baseline, especially with text classification problems due to its efficiency with high-dimensional data.
- Performance: Works well when features are independent (which is a common assumption). If categories are well-separated, Naive Bayes tends to perform decently.
- Confusion Matrix: May show some misclassifications, especially for categories that are similar in content.

Decision Tree Classifier:

- Accuracy: Decision Trees can capture non-linear relationships and interactions between features, potentially leading to higher accuracy than Naive Bayes.
- Performance: Sensitive to overfitting, particularly if the tree depth is not controlled. The model can be complex and may perform well on the training data but poorly on unseen data.
- Confusion Matrix: Can help visualize where the model is making the most errors (specific categories where misclassifications occur).

K-Nearest Neighbors (KNN):

- Accuracy: KNN's performance heavily relies on the choice of k and the distance metric. It may perform well for smaller datasets but can struggle with larger datasets due to increased computation time.
- Performance: KNN can handle multi-class classification but can be affected by noise in the data and the presence of irrelevant features.
- Confusion Matrix: The model might show a pattern where it misclassifies certain categories based on proximity, particularly in cases of closely related articles.

Random Forest Classifier:

- Accuracy: Generally performs very well as it aggregates predictions from multiple trees, which helps reduce overfitting and improves robustness.
- Performance: Handles a large number of features well, making it suitable for text classification tasks. Can also give insights into feature importance.
- Confusion Matrix: Typically exhibits fewer misclassifications compared to the other models, especially if hyperparameters are tuned properly.

General Insights

- Complexity vs. Interpretability: Naive Bayes is simple and interpretable but might not capture complex patterns in the data as effectively as Decision Trees or Random Forests.
- Overfitting Risks: Decision Trees can easily overfit, whereas Random Forest mitigates this by averaging multiple trees.
- Data Characteristics: Performance can vary significantly based on the dataset's nature. If categories have distinct text patterns, ensemble methods like Random Forest and Decision Trees may outperform Naive Bayes and KNN.

Recommendations

- Model Selection: Consider using Random Forest for its strong performance, particularly if the goal is accuracy.
- Hyperparameter Tuning: Explore tuning hyperparameters for Decision Trees and Random Forests to avoid overfitting and improve model generalization.
- Feature Engineering: Enhancing text features (e.g. n-grams, word embeddings) could improve model performance across all algorithms.

Questionnaire:

1. How many news articles are present in the dataset that we have?

```
In [15]: num_articles = len(df)
print(f"Total number of news articles: {num_articles}")
```

Total number of news articles: 2225

2. Most of the news articles are from ____ category.

```
In [16]: most_common_category = df['Category'].value_counts().idxmax()
print(f"Most of the news articles are from the '{most_common_category}' category.")
```

Most of the news articles are from the 'Sports' category.

3. Only ____ no. of articles belong to the 'Technology' category.

```
In [17]: tech_articles_count = df[df['Category'] == 'Technology'].shape[0]
print(f"Only {tech_articles_count} articles belong to the 'Technology' category.")
```

Only 401 articles belong to the 'Technology' category.

4. What are Stop Words and why should they be removed from the text data?

Answer : Stop words are commonly used words (such as 'and', 'the', 'is', etc.) that do not carry significant meaning and can be removed from the text during preprocessing. Removing stop words can help reduce the dimensionality of the data and improve the performance of text classification models by focusing on the more meaningful words.

5. Explain the difference between Stemming and Lemmatization.

Answer : Stemming is the process of reducing words to their base or root form by removing suffixes (e.g., "running" becomes "run"). It may not produce actual words. Lemmatization, on the other hand, reduces words to their base form by considering the context and meaning (e.g., "better" becomes "good"). Lemmatization is typically more accurate than stemming but computationally more intensive.

6. Which of the techniques Bag of Words or TF-IDF is considered to be more efficient than the other?

Answer : TF-IDF (Term Frequency-Inverse Document Frequency) is generally considered more efficient than Bag of Words because it not only

7. What's the shape of train & test data sets after performing a 75:25 split.

```
In [18]: # Define features and target variable
X = df['Article'] # Features (the text of the articles)
y = df['Category'] # Target variable (the categories)

# Now you can perform the train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=42)

# Print the shapes of the train and test sets
print(f"Shape of training data: {X_train.shape}")
print(f"Shape of testing data: {X_test.shape}")
```

Shape of training data: (1668,)

Shape of testing data: (557,)

8. Which of the following is found to be the best performing model..

- Answer : The best performing model can be determined by comparing the accuracy scores and other metrics (e.g., F1-score, precision, recall) from the classification reports. If you ran multiple models, you would typically select the one with the highest accuracy and best performance metrics. For instance:
- "The Random Forest classifier was found to be the best performing model based on accuracy and classification metrics."

a. Random Forest b. Nearest Neighbors c. Naive Bayes

- According to this particular use case, both precision and recall are equally important. (T/F)

Answer : In this case of categorizing news articles, where precision and recall are equally important, it suggests that you want to ensure that the articles are accurately classified into their respective categories (high precision) while also capturing as many relevant articles as possible (high recall).

- Precision measures the proportion of true positive predictions among all positive predictions, indicating how many of the predicted categories were correct.
- Recall measures the proportion of true positives among all actual positive instances, indicating how well the model can identify all relevant articles.

Balancing both metrics is crucial in multi-class classification tasks like this one to avoid missing relevant articles and to minimize false positives. You can also consider using the F1-score, which is the harmonic mean of precision and recall, to evaluate the model performance when both metrics are important.